

Diploma in Operation Theater Technology

SCHEME OF EXAMINATION

Subject code	Title of the Course	Hours	Passing Maximum
Semester I			
Subject 1	Anatomy and physiology	3 Hours	40/100
Subject 2	Fundamentals of Health	3 Hours	40/100
Subject 3	Medical Surgical Care- I	3 Hours	40/100
Subject 4	Communicative English	3 Hours	40/100
Subject 5	Practical – I - Basic Care	3 Hours	40/100
Semester II			
Subject 1	Operation Theatre Techniques	3 Hours	40/100
Subject 2	Medical Surgical Care- II	3 Hours	40/100
Subject 3	Life Skill	3 Hours	40/100
Subject 4	Practical II- Positioning and Operating Room Procedures	3 Hours	40/100
Subject 5	Project– Advanced Care	3 Hours	40/100
Semester III			
Subject 1	Pharmacology	3 Hours	40/100
Subject 2	Microbiology	3 Hours	40/100
Subject 3	General Surgical Procedures	3 Hours	40/100
Subject 4	Practical III-Internship Assisting Minor operations	3 Hours	40/100
Subject 5	Practical IV-Project work – Minor operative procedures	3 Hours	40/100
Semester IV			
Subject 1	Sociology and Patient Rights to Ethics	3 Hours	40/100
Subject 2	Anaesthesia	3 Hours	40/100
Subject 3	Perioperative Care	3 Hours	40/100
Subject 4	Practical V – Assisting Major operations	3 Hours	40/100
Subject 5	Project/Internship	3 Hours	40/100

Eligibility for admission: Pass in 12th std Examination conducted by the Govt. of Tamil Nadu Board of Secondary Education, Government of Tamil Nadu or any other equivalent Examination.

Examination: Passing minimum for each Course is 40%.classification will be done on the basis percentage marks of the total marks obtained in all of the Courses and given as below:

- | | | |
|-----------------------|---|--------------|
| 40% but less than 50% | - | Third Class |
| 50%but less than 60% | - | Second Class |
| 60% and above | - | First Class |

Theory Paper

Internal Marks-25

External Marks-75

SYLLABUS**SEMESTER I**

Subject 1	:	Anatomy and physiology
Subject 2	:	Fundamentals of Health
Subject 3	:	Medical Surgical Care- I
Subject 4	:	Communicative English
Subject 5	:	Practical I - Basic Care

SEMESTER II

Subject 1	:	Operation Theatre Techniques
Subject 2	:	Medical Surgical Care- II
Subject 3	:	Life Skill
Subject 4	:	Practical II- Positioning and Operating Room Procedures
Subject 5	:	Project- Advanced Care

SEMESTER III

Subject 1	:	Pharmacology
Subject 2	:	Microbiology
Subject 3	:	General Surgical Procedures
Subject 4	:	Practical III - Internship Assisting Minor operations
Subject 5	:	Practical IV-Project work-Minor operative procedures

SEMESTER IV

Subject 1	:	Sociology and Patient Rights to Ethics
Subject 2	:	Anaesthesia
Subject 3	:	Perioperative Care
Subject 4	:	Practical V - Assisting Major operations
Subject 5	:	Project/Internship

Program Objectives

1. Communicate effectively using interpersonal skills
2. Utilize evidence-based interventions to achieve optimal adaptation.
3. Safely perform and prioritize appropriate PRE-INTRA-POST OPERATIVE care skills.
4. Employ appropriate thinking to safely and effectively PERFORM OPERATIVE ROOM TECHNIQUES

SEMESTER I

SUBJECT I

ANATOMY AND PHYSIOLOGY

Objectives

- To equip the students with the knowledge of human body system.
- To understand the function of various structure of the human body

Unit I

18 Hrs

The Body as a whole - Cavities of the body and their Contents – Cells and tissues – Organs and systems – The skin and appendages – Structure and function of skeleton - The skull – The Vertebral column – The Thorax – Bones of the upper limbs – Bones of the lower limbs. Joints and Muscles – Types of Joints – The muscular system.

UNIT II

18 Hrs

Nervous system- The brain – cerebrum – cerebellum – mid brain – brain stem – pons and Medulla. The spinal cord and spinal nerves - The meninges and cerebrospinal fluid. The eye – The ear – The sense of smell and taste

UNIT III

18 Hrs

Circulatory system -The blood – Blood clotting – Blood grouping – Cross matching - The heart and blood vessels. Respiratory System-The nose pharynx – larynx – The Trachea – Bronchi – Lungs.

Unit IV

18 Hrs

Digestive System – structure and functions-The mouth – pharynx – Oesophagus – The stomach – small intestine- large intestine –rectum-anus- liver-salivary glands-gall bladder-pancreas

Unit V

18 Hrs

The Endocrine glands – Thyroid – Parathyroid – Pituitary – Adrenal gland. Male and female reproductive system,- The Excretory system –Kidneys-ureters-urinary bladder-urethra- Urine and micturation - water and salt balance

SUBJECT II

Fundamentals of Health

Objectives

- The student know about assessment of patient, care of patient ,basic needs and maintaining health.
- To improve skill to taking care of patient

Unit I

18 Hrs

Vital signs-temperature-pulse-respiration – Blood pressure – weight checking – height checking – history taking –methods of physical examination-types of positions-preparation for diagnostic procedures

Unit II

18 Hrs

Drugs-uses-sources-classification of drugs-routes of administration-oral medication – injection – infusion – hot and cold application –dressing of a wound— Types of record – types of Reports.

Unit III

18 Hrs

Hand washing technique – use of gloves – use of masks in medical asepsis - mask – gloves – gowns for surgical use – surgical hand scrub – handling the sterile articles-admission of patient - discharge of patient-death care.

Unit IV

18 Hrs

Care of the skin – Care of the hair –pediculosis treatment Care of the mouth – Care of pressure sore – meeting eliminational needs –urine test-collection of specimen- bed making - Use of comfort device.

Unit V

18 Hrs

Enema – bowel wash – oxygen administration – ryles tube feeding – Catheterization-nebulizer-assisting for vaginal –rectal examination-lumbar puncture-pre and post operative care.

SUBJECT III

Medical and Surgical Care - I

OBJECTIVES

- To Have a Knowledge Of Medical Surgical Nursing
- To Understand The Various Problems And Treatment Of Diseases

Unit I

18 Hrs

Skeletal system including, diseases, risk factors, signs and symptoms and diagnosis and treatment-Osteoarthritis.Rheumatoid Arthritis-Arthritis- Lateral Epicondylitis (Tennis Elbow) Medial Epicondylitis (Golfer's or Baseball Elbow)-Carpal Tunnel Syndrome-Ligament Injuries to the Knee.

Unit II

18 Hrs

Cardiology system including diseases, risk factors, signs and symptoms, and diagnosis and treatment-Coronary artery disease-Atherosclerosis-Hypertension.-Heart failure-Aortic aneurysm-Myocarditis and pericarditis.-Cardiomyopathy.

Unit III

18 Hrs

Urinary system including diseases, risk factors, signs and symptoms ,and diagnosis and treatment-Bladder Cancer-Bladder Diseases.-Interstitial Cystitis-ureter and Kidney Stones-Urinary Incontinence.

Unit IV

18 Hrs

Respiratory system including diseases, risk factors, signs and symptoms and diagnosis and treatment-Chronic Obstructive Pulmonary Disease (COPD) - Chronic Bronchitis. -Emphysema-asthma-Lung Cancer-Bronchiectasis. - Pneumonia.-Pleural Effusion.

Unit V

18 Hrs

Reproductive diseases, risk factors, signs and symptoms, diagnosis and treatment-Prostate cancer - Breast cancer - Ovarian cancer - Cancer of the ovary-infertility-fibroid uterus-Uterine cancer - Testicular cancer.

SUBJECT IV

Communicative English

1. Basic Grammar:

- a. Review of grammar
- b. Remedial study of grammar
- c. Simple sentence
- d. Word passive voice etc.

2. Bubbling Vocabulary:

- a. Synonyms
- b. Antonyms
- c. One – word Institution

3. Reading and Understanding English

- a. Comprehension passage
- b. Précis – writing
- c. Developing a story from hints.

4. Writing English

- a. Writing Business letters.
- b. Paragraph writing
- c. Essay writing
- d. Dialogue writing

5. Speaking English

- a. Expressions used under different circumstances
- b. Phonetics

SUBJECT V
Practical I
Basic Care

- | | |
|----------------------------|------------------------------------|
| a) The surgical team | g) Glass ware |
| b) Prevention of infection | h) Drugs and solutions |
| c) Hazards | i) Linen |
| d) Personal attitude | j) Admission of patient to theatre |
| e) Equipment | k) Draping |
| f) Rubber goods | l) Care of specimen |

SEMESTER II
SUBJECT VI

Operating Room Techniques

Objectives

- The student gain more skill about Operating Room Techniques

Unit I **18 Hrs**

Infection and Control Including Asepsis And disinfection, Hospital Waste Management. Role Of Nurses In Infection control-surgical hand washing, mask, gown, glove using techniques

Unit II **18 Hrs**

Responsibility of scrub person and circulating person General operative procedures equipments and uses-laparotomy, minor set, major set, gynaecological ,orthopaedic ,thoracic instruments

Unit III **18 Hrs**

Blood transfusion-Antigen antibody reactions ,hypersensitivity-Sutures-types-needles-types-blades -haemostatic agents.

Unit IV **18 Hrs**

Pre operative check list informed consent-Patient identification -safe surgery check list - allergy- any infectious disease-site marking-surgical count-Handling sterile equipment and autoclave

Unit V **18 Hrs**

Management of emergency equipment-Oxygen cylinder-suction apparatus-diathermy-defibrillator-positioning and safety measures-equipment know how and maintenance of operating table-dome light-scopes

SUBJECT VII
Medical and Surgical Care II

Objectives

- To Have a Knowledge Of Medical Surgical Nursing
- To Understand The Various Problems And Treatment Of Diseases

Unit I

18 Hrs

Gastro Intestinal System Including Diseases And Risk Factors, Signs And Symptoms, Diagnosis And Treatment-Gastroesophageal reflux disease-Appendicitis-peptic ulcer-intestinal obstruction-hernia

Unit II

18 Hrs

Brain System Including Sign And System Diagnosis Treatment-Alzheimer's disease. Bell's palsy-cerebral palsy.-Epilepsy. -Motor neurone disease (MND) -Multiple sclerosis (MS) -Parkinson's disease.

Unit III

18 Hrs

Immune system disease including care and treatment-Systemic lupus erythematosus (lupus)-)-Multiple myeloma-Type 1 diabetes mellitus-Guillain-Barre syndrome-hodgkin lymphoma

Unit IV

18 Hrs

ENT-diseases and surgeries -Dizziness-Dysphagia (Difficulty Swallowing)Ear Infection (Otitis Media-Hearing Loss-Hoarseness-Meniere's-Nosebleeding--Sleep Apnea-Snoring-Tinnitus (Ringing in the Ears)-TMJ Pain (Temporo-Mandibular Joint)-Tonsils & Adenoid Problems

Unit V

18 Hrs

EYE -diseases and surgeries Age-related Macular Degeneration (AMD) of eye-. -Cataract-Diabetic Eye Disease-Glaucoma.-Dry Eye.-Low Vision-Ectropion. -Entropion.-Facial palsy-Trichiasis.

SUBJECT VIII

Life Skill

I Life Coping or adjustment

- (a) External and internal influence in one's life
- (b) Process of coping or adjustment
- (c) Coping with physical change and sexuality
- (d) Coping with stress, shyness, fear, anger far live and criticism.

II Attitude

- (a) Attitude
- (b) Self acceptance, self – esteem and self actualization
- (c) Positive thinking

III Problem Solving

- (a) Goal Setting
- (b) Decision Making
- (c) Time Management and stress Management.

IV Computers

- (a) Introduction to Computers
- (b) M.S.Office
- (c) Power Point

V Internet

- (a) Introduction to internet
- (b) E – mail
- (c) Browsing

SUBJECT IX

Practical II

Positioning and Operating Room Procedures

- Identification of Instruments and its uses
- Sterilization -Autoclave
- Suture material
- Set up of suction apparatus
- Diathermy
- Responsibility to anaesthetist
- Minor procedures
- Position of patient
- Operating instruments trolley set up
- Assisting for procedures-pre-intra-post
- documentation

SUBJECT X

Project

Advanced care

Advanced care in

Sterilization -Autoclave -CSSD

LINEN AND EQUIPMENT SET UP IN STERILE TRAYS

- Set up of suction apparatus
- Diathermy
- Operating instruments trolley set up
- Minor procedures

Semester III
SUBJECT XI
Pharmacology

Objectives

- The student know about the indication ,contra indication ,dosage ,side effects of drugs.

Unit I

18 Hrs

Medications used in operating room.Medications actions and side effects-adversereaction

Unit II

18 Hrs

Emergency Drugs –drug calculation-formula-Adrenaline : Mode or administration, dilution, dosage,-Effects, Isoprenaline-Atropine, bicarbonate, calcium, ephedrine, xylocard,-Inotropes : dopamine, dobutamine, amidaron-Aminophylline, hydrocortisone, antihistamines, potassium.-Cardiovascular drugs-Antihypertensives-Antiarhythmics-Beta Blockers-Ca - Channel blockers.

Unit III

18 Hrs

Vasodilators - nitroglycerin& sodium nitroprusside- Respiratory system - Bronchodilators-respiratory stimulants -Broncholytic agents -Renal-system-Diuretics,furosemide,mannitol-Obstetrics - oxytocin,methergin-Miscellaneous - Antibiotics NSAIDs Anticoagulants and Insulin-IV Solutions And Administration

Unit IV

18 Hrs

Atropine, glycopyrrolate-Sedative-anxiolytics-Diazepam, midazolam, phenergan, lorazepam, chlorpromazine,-Trichlopho-Narcotics-Morphine, pethidine, fentanyl, pentazocine-Antiemetics-Metoclopramide,ondansetron, dexamethasone-Antacids-Na citrate, gelusil, mucaine gel.-H2 blockers-Cimetidine, ranitidine, famotidine

Unit V

18 Hrs

Anesthetic drugs and administration-Induction agent-Thiopentone , diazepam, midazolam, ketamine, propofol, etomidate.-Muscle relaxants-Depolarising -suxamethonium,-Non depolarising -pancuronium, vecuronium, atracurium, rocuranium

SUBJECT XII

Microbiology

Objectives

- To discuss the characteristics of micro organisms including bacterial,viral and fungal infections
- To discuss the measures to control the spread of micro organisms

Unit I

18 Hrs

Micro organism – classification of micro organism and their characteristics – pathogenic micro organism – paths of infection – portal of entry – portal of exit.

Unit II

18 Hrs

Disinfection – methods of disinfection – sterilization – methods of sterilization – natural agents - physical agents – chemical agents.

Unit III

18 Hrs

collection specimen – blood, urine, stool, sputum, discharges.

Unit IV

18 Hrs

Immunity – types of immunity – Immunoglobulin, vaccines, types of vaccines, national immunization schedule.

Unit V

18 Hrs

Food poisoning – blood borne infection – microscope – types of microscope – uses of microscope.

SUBJECT XIII
General surgical procedures

Objectives

- The student gain more skill about surgical procedures

Unit I

18 Hrs

Laparotomy and gynaecology procedures-Adhesiolysis-Cervical (Cone) Biopsy
Colporrhaphy-Colposcopy-Dilation and Curettage (D&C) Endometrial Ablation-
Endometrial or Uterine Biopsy-Hysterectomy-Tubal ligation

Unit II

18 Hrs

Surgical management of Fracture-Joint Replacement Procedures-knee-total hip
replacement-Revision Joint Surgery. -Debridement--Bone Fusion-External and
Internal Fixation of Bones.-Osteotomy.

Unit III

18 Hrs

Arthroscopy-laryngoscopy-Bronchoscopy- lungs-Colonoscopy and sigmoidoscopy:
large intestine.-Cystoscopy-ureteroscopy.-Laparoscopy: abdomen or pelvis-Upper
gastrointestinal endoscopy: oesophagus and stomach.

Unit IV

18 Hrs

Minor surgeries

Tonsillectomy-nasal polyp removal-Breast Biopsy.-Burn Excision/Debridement.-
Cataract Surgery.-Caesarean Section.-Circumcision.-Dental Restoration.Gastric
Bypass.

Unit V

18 Hrs

Major surgeries- organ transplant-Craniotomy-removal of a brain tumour,
nephrectomy-open-heart surgery-valve replacement-thoracotomy-pneumonectomy

SUBJECT XIV
Practical III-Internship

ASSISTING FOR MINOR OPERATIONS

- INCISION AND DRAINAGE
- DILATATION AND CURETTAGE
- APPENDICECTOMY
- INGUINAL HERNIA REPAIR
- TONSILLECTOMY
- CIRCUMCISION

SUBJECT XV
Practical IV

PROJECT WORK- Minor Operative Procedures

Assessment of patient

- Assisting for Anaesthesia
- Pre Operative Care
- Intra operative assessment
- Immediate Post Operative Care

SEMESTER IV
SUBJECT XVI
Sociology and Patient Rights to Ethics

Objectives

- The student know about the patient rights and take care of patient
- Understand follow up the medical ethics and social behaviour.

Unit I

18 Hrs

Understanding and learning about community-groups-the family-urban and rural administrative pattern- corporation, .panchayats and co-operatives -social processes-social controls

Unit II

18 Hrs

Social stratification and leadership patterns in the community-the community and factor affecting its health and development-gathering information about our communities

Unit III

18 Hrs

Patients Rights , Public Health Ethics Detail Document About Patient Ethics - Common Principles of Nursing

Unit IV

18 Hrs

Medical ethics - Definition - Goal - Scope-Code of conduct - Introduction -Basic principles of medical ethics – Confidentiality-Malpractice and negligence - Rational and irrational drug therapy

UNIT V

18 Hrs

Autonomy and informed consent - Right of patient- Care of the terminally ill-Organ transplantation

SUBJECT XVII

Anaesthesia

Objectives:

- The student gain more skill about Anaesthetic Procedure

Unit I

18 Hrs

PRE-OP PREPARATION:

Pre anaesthetic assessment~ History – past history - disease - Surgery and personal history - Smoking / alcohol General physical assessment, systemic-examination – CVS, RS, CNS

INTRODUCTION TO GENERAL ANAESTHESIA

INHALATIONAL GASES-Gases - O₂, N₂O, Air-Agents - Ether-, Halothane, Isoflurane, Sevoflurane, Desflurane

REVERSAL AGENTS-Neostigmine, Glycopyrrolate, Atropine, Nalorphine, Naloxone, Flumazenil (Diazepam)

Unit II

18 Hrs

ANAESTHESIA MACHINE

Hanger And Yoke System-Cylinder Pressure Gauge- Pressure Regulator Flow Meter Assembly-Vapourizers - Types, Hazards, Maintenance, Filling And Draining, Etc.- Face Masks & Airway Laryngoscopes- Types, Sizes Endotracheal Tubes - Types, Sizes.- Cuff System-Fixing, Removing And Inflating Cuff, Checking Tube Position Complications MONITORING-ECG-SP02-TEMPERATURE-IBP-CVP-PA-PRESSURE-LA Pressure

Unit III

18 Hrs

Bio Medical engineering of Trouble sorting Management, care of cleaning LOCAL ANAESTHETICS-Xylocaine, Preparation, Local – Bupivacaine - Topical, Prilocaine-jelly, Emla - Ointment, Etidocaine.Ropivacaine

Unit IV

18 Hrs

Principles Of Anaesthesia -Basic Anesthetic Techniques PRE - ANAESTHETIC ORDERS:-Patient - Informed consent- Npo guidelines - Premedication - advantages, drugs used- Special instructions - if any Machine - Checking the machine-O₂, N₂O, suction apparatus-Laryngoscopes-ET tubes, airways- Things for IV accessibility

UNIT V

18 Hrs

-INTRAOPERATIVE MANAGEMENT

Confirm the identification of the patient- Monitoring - minimum-Noninvasive& Invasive monitoring- Induction - drugs used- Endotracheal intubation Maintenance of anaesthesia- Positioning of the patient-Blood / fluid & electrolyte balance-Reversal from anaesthesia - drugs used- Transferring the patient-Recovery room – set up and things needed Other monitoring systems

SUBJECT XVIII
Perioperative Care

Objectives

- To know the student about how to care the perioperative care patient.
- To gain more knowledge about perioperative procedure

Unit I

18 Hrs

Preoperative care-information to operating room-skin preparation-blood arrangements-lab investigations-Assessment, planning, implementation and evaluation of care.

Unit II

18 Hrs

Assessment, planning, implementation and evaluation of care on the day of surgery-informed consent-Patient identification -safe surgery check list - allergy-any infectious disease-pre op medications

UNIT III

18 Hrs

General surgical procedures- -orthopaedics-paediatric -gynaecological-obstretical-eye-ENT-thoracic surgery- Instruments in emergency trolley.

Unit IV

18 Hrs

Common surgical terminologies-ectomy, otomy, ostomy Types of anaesthesia-.general-local-spinal-types of surgery- Elective surgery -Emergency-minor-major-reconstructive surgery

Unit V

18 Hrs

Post operative care- immediate and special care-Prevention of post operative complications --shock- Bleeding-Wound infection-Deep vein thrombosis-Pulmonary embolism-.Urinary retention.-.Reaction to anaesthesia.

SUBJECT XIX

Practical V ASSISTING MAJOR OPERATIONS

- LSCS
- LAPAROTOMY
- THYROIDECTOMY
- HYSTRECECTOMY
- OPEN REDUCTION AND INTERNAL FIXATION

SUBJECT XX Project/Internship

INCLUDING

- Aseptic Techniques
- Ventilation
- Mechanical Intubation
- Safety Measures
- Documentation
